ATHUL KRISHNA K M

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PROFESSIONAL SUMMARY

Detail-oriented **Junior Robotics Engineer** with hands-on experience in **ROS 2 control systems, robotic manipulator integration, and teleoperation setups**. Skilled in developing **C++ hardware interfaces, MoveIt 2 configurations, and web-based remote visualization frameworks**. Proactive and adaptable, with a strong background in Computer Science, specialized in bridging physical hardware with simulation environments to deliver reliable, production-ready robotic applications.

WORK EXPERIENCE

Junior Robotics Engineer | I Hub Robotics, Aluva, Ernakulam — June 2026 – Present

- Spearheading development of production-grade robotic control interfaces and hardware integration pipelines.
- Implementing manipulation configurations using MoveIt 2 and inverse kinematics (IK) solvers for LeRobot manipulator platforms.
- Deploying and calibrating real-time leader-follower mimicking systems for synchronized robot arm control.
- Assembling, calibrating, and configuring navigation and SLAM setups for TortoiseBot mobile platforms.

Robotics Engineer Intern | I Hub Robotics, Aluva, Ernakulam — April 2026 – June 2026

- Developed a complete ROS 2 control architecture (ROS 2 Humble) for the 4-DOF Rotrics DexArm manipulator, integration and custom C++ hardware interfaces.
- Designed and implemented a C++ and Python-based Teach-and-Play (Teach-and-Replay) system featuring pose recording, relative motion mapping, encoder feedback, and emergency stops.
- Researched and configured leader-follower servo setups and relative motion mapping for teleoperation on the LeRobot platform.
- Built a web-based ROS 2 visualization dashboard with HLS video streaming (Python/FFmpeg) and remote topic subscription over secure Cloudflare tunnels.

PROJECTS

Autonomous Hospital Delivery Robot (Med-Bot) — 2026

- Developed a complete simulation-based autonomous medicine delivery robot using ROS 2 Humble, Gazebo, and Nav2, utilizing a differential drive setup and ros2-control.
- Implemented a custom delivery manager node in Python for multi-room route optimization, continuous logging, and automated return-to-pharmacy utilizing precise AMCL localization.
- Designed a professional PyQt5 control dashboard for real-time monitoring, integrating an emergency stop system and twist-mux for seamless manual teleoperation override.

Attendance Marking System Using Face Recognition | Python, OpenCV, HTML/CSS — 2025

- Developed a face recognition-based attendance system that automates student attendance using real-time camera input.
- Implemented image training using pre-trained face recognition model and executed real-time recognition and attendance logging using Python.
- Created a simple front-end interface using HTML/CSS for user interaction and display of attendance data.
- Utilized OpenCV for facial detection and recognition, reducing manual errors and improving efficiency in attendance management.

TECHNICAL SKILLS

Robotics & ROS 2:

- ROS 2 (Humble), URDF/Xacro modeling, MoveIt 2, RViz2, Gazebo, SLAM, Nav2
- Hardware Integration: DexArm, LeRobot (leader-follower mimicking), TortoiseBot platforms

Programming & Tools:

- Python, C++, C, Java, HTML/CSS, OpenCV
- Linux (Ubuntu), Bash, Git, VS Code

CERTIFICATIONS AND WORKSHOPS

- OPENCV BOOTCAMP by OpenCV University, January 2026
- Workshop on Artificial Intelligence with Python by Techmaghi, April 2024

EDUCATION

IHub School of Learning, Edappally, Kerala, India September 2025 – Present

- PG Diploma in Robotics Operating System

College Of Engineering Munnar, Kerala, India June 2021 – April 2025

- B.Tech in Computer Science